

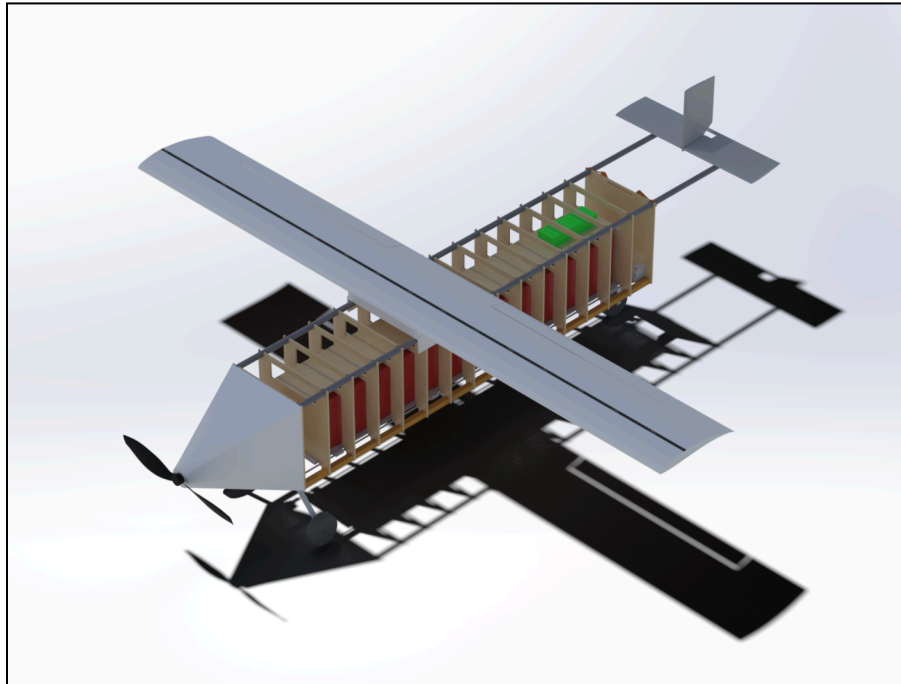
MAE 155B Team 3: Citadel

Detail Design Review

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1. Executive Summary

1.1 Background and Motivation

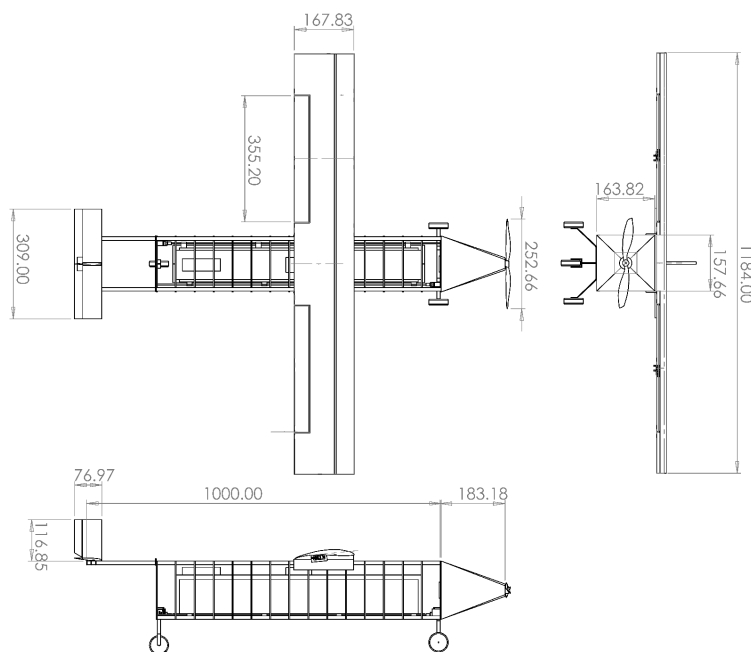
Our design, achieved through a profit sensitivity analysis and design optimization, prioritizes the volume payload size with secondary emphasis on minimizing flight time and maximizing the weight payload. Our design employs a conventional tube-and-wing aircraft architecture with a fuselage length of 0.8 meters, capable of holding a volume payload of about 6.25 liters. A twin-bay door deployment system is implemented to meet the mission requirement for mid-flight volume payload deployment, with special consideration for the large volume we aim to deploy. In addition to the volume payload, the aircraft is equipped to carry a 5.77 N weight payload. The center of gravity (CG) for both payloads is aligned with the CG of the aircraft.

Winged surfaces are made of XPS foam to ensure ease of manufacture and weight savings, and reinforced spanwise using carbon fiber spars. The fuselage utilizes a built-up wooden structure, with 14 equidistant wooden ribs placed along its length, and two firewalls at either end. The wooden ribs are connected by two carbon-fiber rods at the upper corners and two spruce spars at the bottom corners to ensure structural integrity. The carbon-fiber rods extend behind the fuselage and connect to the empennage of our aircraft. A 3D printed nosecone sits at the front, connecting the fuselage to the 10x7E propeller and the motor.

Table 1.1.1: Key design parameters of our profit-maximizing RC aircraft.

Gross Weight [N]	Payload 1 Weight [N]	Payload 2 Volume [L]	Wingspan [m]	Cruise L/D	Propeller Size	W/S [N/m ²]	T/W
20.02	5.77	6.234	1.184	8.39	10x7	100	0.19

1.2 Technical Drawing (1:10 scale, dimensions in mm)



2. Preliminary Aircraft Design

2.1 Profit Function Analysis & Multidisciplinary Design Optimization

A Monte Carlo simulation is employed to determine the set of most influential scoring function parameters. Bounded by reference designs from previous competition years' aircraft, the design space is sampled $1e9$ times, revealing through a Spearman Rank Correlation that V_p and T_f influence the profitability of the aircraft significantly more than other parameters. A visualization of these findings is available in Appendix 8.2.1.

Gradient-based Multidisciplinary Design Optimization (MDO) is employed to balance sizing and performance parameters to maximize profitability. The solver is implemented in Python utilizing JAX for automatic gradient computation, modOpt for problem formulation, and IPOPT as the nonlinear algorithm. Compared to gradient-free approaches, gradient-based approaches significantly improve convergence accuracy and solver efficiency.

The following design parameters are considered: $[W_p, V_p, V_{cruise}, S_{wing}, S_{ht}, S_{vt}, X_{wing_frac}, X_{total}, X_{battery_frac}]$, and all are normalized to $O(1)$ to prevent ill-conditioning. The objective $J(x)$ is defined to maximize profit (\$/hr). A total of 24 constraints are applied across five categories to ensure feasibility: aerodynamic performance (thrust, stall, climb, maneuver), geometric bounds (wing area, gross weight, wing loading), electrical (motor current), stability (static margin, tail volume coefficients), and physical packaging (landing gear tip-back, payload placement, battery overlap, aircraft length). The following eight coupled physics models are evaluated for each design: 3D lift and drag, thrust with speed and fuselage knockdown corrections, power consumption, neutral point, aircraft weight and center-of-gravity estimation, tail volume coefficients, motor current, and mission length.

The MDO results reveal several design tradeoffs that traditional design methods may not detect. The low L/D of 8.39 and the respective shallow cruise angle of 0.49° suggest that, compared to full-scale aircraft, aerodynamic efficiency is traded for profit, where higher drag is accepted over a smaller, lighter, more efficient airframe. The optimal battery CG location at 95% of fuselage length highlights its role as the primary mechanism to shift the CG closer to the neutral point to ensure static stability. The cruise speed of 19.6 m/s - two-thirds of the upper bound - indicates that beyond this point, gains in flight time are outweighed by rising energy costs and structural loads, making faster flight counterproductive to profit.

2.2 Key MDO Results

Aircraft Weight:	$W_g = 20.02 \text{ N} \mid W_e = 11.66 \text{ N} \mid W_p = 6.616 \text{ N} \mid W/S = 100.00 \text{ N/m}^2$
Payload & Fuse:	$V_p = 5.347 \text{ L} \mid L_p = 67.36 \text{ cm} \mid W_p = 9.62 \text{ cm} \mid L_f = 1.050 \text{ m} \mid W_f = 16.46 \text{ cm}$
Geometry:	$S = 0.2002 \text{ m}^2 \mid b = 1.184 \text{ m} \mid MAC = 0.1691 \text{ m} \mid S_{ht} = 0.0128 \text{ m}^2 \mid S_{vt} = 0.0095 \text{ m}^2$
Stability:	$X_{cg} = 51.1\% \mid X_{np} = 53.5\% \mid SM = 15.00\% \mid C_h = 0.300 \mid C_v = 0.0322$
Aero:	$\alpha = 0.49^\circ \mid C_L = 0.4256 \mid C_{D0} = 0.04044 \mid C_D = 0.05119 \mid L/D = 8.39 \mid D = 2.408 \text{ N}$
Performance:	$V = 19.59 \text{ m/s} \mid V_{stall} = 9.15 \text{ m/s} \mid V_{climb} = 10.66 \text{ m/s} \mid T_f = 63.1 \text{ s} (1.05 \text{ min})$
Propulsion:	$T = 3.799 \text{ N} \mid T/W = 0.190 \mid I = 35.00 \text{ A} \mid E_f = 4252.3 \text{ J} (1.1812 \text{ Wh})$
Profit:	$J(x) = 0.001536 \text{ \$/s} (5.5308 \text{ \$/hr})$

3. Propulsion

Our RC aircraft uses a 10×7E Advanced Precision Composites (APC) propeller, selected based on compatibility with the SunnySky X2216 V3 brushless motor and preliminary aircraft design. A steady-state motor model was used to define motor torque, while APC propeller data [R3] was used to construct 2D surrogate models for thrust and torque as functions of rotational speed and advance ratio for each candidate propeller. The torque balance equation was solved at the aircraft’s cruise condition, and propellers requiring excessive current were eliminated. Further selection prioritized thrust generation over efficiency, as payload capacity is the dominant factor in driving profit. Thrust was determined from flight constraint analysis, and among the viable options, the 10×7E propeller provided the highest thrust, including cruise and fuselage knockdown factors. Key performance properties are summarized in Appendix 8.2.6, and the variation of thrust coefficient with cruise speed is shown in Appendix 8.2.5, demonstrating that the propeller meets cruise performance requirements.

4. Aerodynamics

The main wing uses a high-lift S1223 airfoil optimized using Neural Foil to maximize lift and aerodynamic efficiency for the expected operating angles of attack. The wing profile features a 12.1% thickness-to-chord (t/c) ratio, where the maximum thickness is at 7.9% of the chord length. The optimized airfoil is evaluated in 2D using XFOIL for a calculated Reynolds number of 184,000. The resulting 2D aerodynamic parameters are then extrapolated to 3D aerodynamic coefficients of the aircraft using the semi-empirical lift curve slope method coupled with Prandtl’s lifting line correction. The contributions of the main wing and empennage, as well as the fuselage, propeller, and landing gear, are all accounted for.

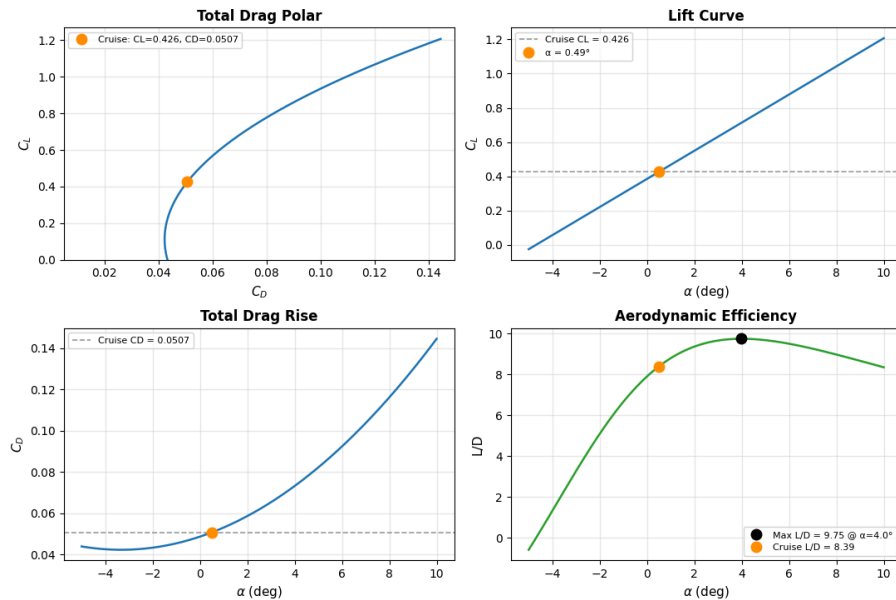


Figure 4.1.1: 3D aircraft aerodynamic performance for the optimized S1223 through Neuralfoil.

5. Weight Distribution and Stability

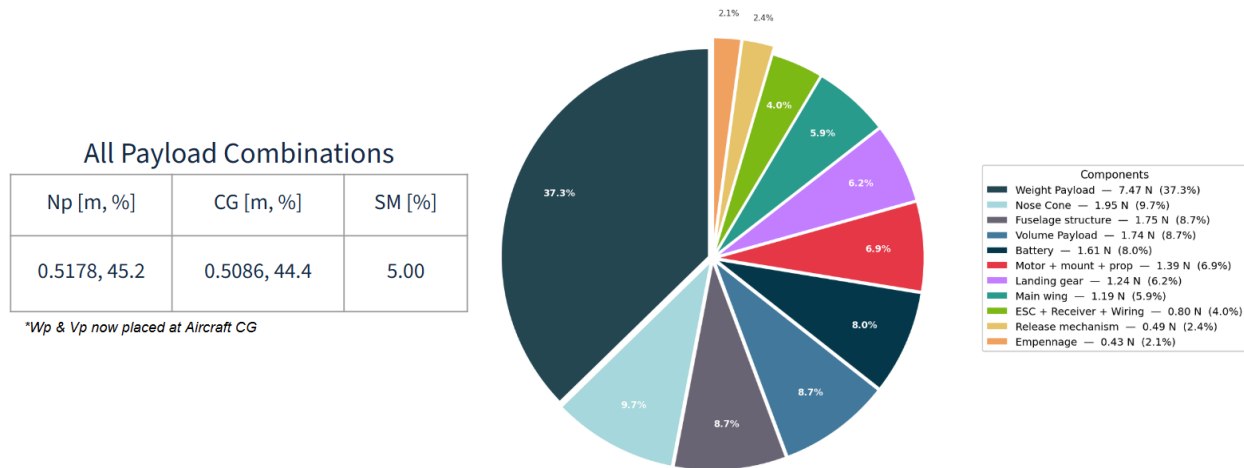


Figure 5.1.1: Weight distribution across key aircraft components

The weight payload and the volume payload take up a substantial portion of the aircraft's weight. In our design and in the manufacturing process, special care is taken to ensure that their CGs align with that of the entire aircraft, such that adverse moments are not created. The 3D-printed nose cone also contributes significantly to the total weight. While this increases the forward mass of the aircraft, it can be beneficial because the aircraft requires sufficient nose weight to balance the empennage and aft structural components (which are often underestimated before build). The battery is treated as an adjustable mass within the fuselage, allowing its longitudinal position to be tuned during final assembly to achieve the desired center-of-gravity location and static margin.

6. Main Wing and Empennage

The highly cambered nature of the S1223 airfoil motivates the material choice of XPS foam to ensure ease of manufacturability (8.3.1). To account for aerodynamic loads during flight, the main wing is supported by two spanwise carbon fiber spars (8.3.2) at the upper and lower surfaces at maximum t/c. Using the female foam mold from the cutout, the wingbox is epoxied flush to the bottom surface of the wing with a span matching the fuselage width, acting as a wing saddle (8.2.2). Balsa rectangles epoxied onto the foam hug the fuselage to fix the wing position laterally. To provide longitudinal stability, the balsa rectangular cut-outs have notches that slide around the built-up balsa ribs of the fuselage. The wing's ailerons take up 30% of the wingspan and are joined to the wing using electric flying hinge tape. To control each aileron, two independent servo motors (8.3.3) are attached to the underside of the wing using plywood mounts (8.3.4) and a servo linkage allowing for 20 degrees of motion.

The vertical (8.3.5) and horizontal (8.3.6) stabilizers are cut from XPS foam. The horizontal stabilizer has a cutout at its center line to interface with the vertical stabilizer, ensuring an angle of attack of -4 degrees and adequate room for elevator and rudder movement. Carbon fiber rods connecting fuselage ribs on the upper surface extend out past the fuselage to serve as twin tail booms. They connect to the empennage via custom-built balsa ribs that are epoxied to the bottom of the vertical stabilizer extension. Servo motors

(8.3.3) to actuate control surfaces are housed at the rear of the fuselage and connect to the empennage through wires. A balsa cutout rests on the tail booms to ensure wires are fed properly to control surfaces. Control surfaces take up 30% of the wingspan and are joined to the wing using electric flying hinge tape.

7. Fuselage

7.1 Fuselage Construction

The fuselage utilizes a built-up wooden structure, with 14 wooden ribs placed equidistantly along its length (8.3.8), and two firewalls (8.3.11) at either end. The wooden ribs are supported by two carbon-fiber rods at the upper corners and two L-shaped spruce spars (8.3.10) at the bottom corners spanning the entire fuselage to ensure structural integrity and prevent buckling. The two firewalls are made using plywood to ensure structural integrity during periods of high loading. To reduce drag effects during flight, a 3D printed nose cone (8.3.7) sits at the front of the fuselage, connecting it to the 10x7E propeller and the motor. This connection is made possible by four wooden spars, which are epoxied to a plywood mount on the front firewall while also being epoxied to the motor mount housed inside the hollow nose cone.

The wooden ribs across the main fuselage are cut out to divide the fuselage into two sections: 1) the bottom section, which houses the 6.25-liter volume payload, and 2) the top section, which houses the weight payload, battery, electronics, and all wiring. The wooden rib cutouts are wrapped with Monokote to improve the aerodynamics of the exterior. The rear of the fuselage houses two servo motors, which actuate the control surfaces of the horizontal and vertical stabilizers. For seamless integration with the empennage, the two carbon fiber spars along the upper corners extend beyond the end of the fuselage and into the tail-boom box, as discussed in section 4.2. The bottom surface of the fuselage is simply the bay doors of the release mechanism.

7.2 Release Mechanism

The payload release mechanism is responsible for deploying our high-volume payload (6.25 L) and utilizes a twin bay door configuration. Each bay door (8.3.12) is connected to a rotating actuation rod (8.3.13) driven by a dedicated servo, resulting in a total of two servo motors (8.3.3) for the entire system. Torque is directly transferred from the servo to the rod through a 3D printed attachment (8.3.14), transmitting motion to the bay doors for package deployment and retraction.

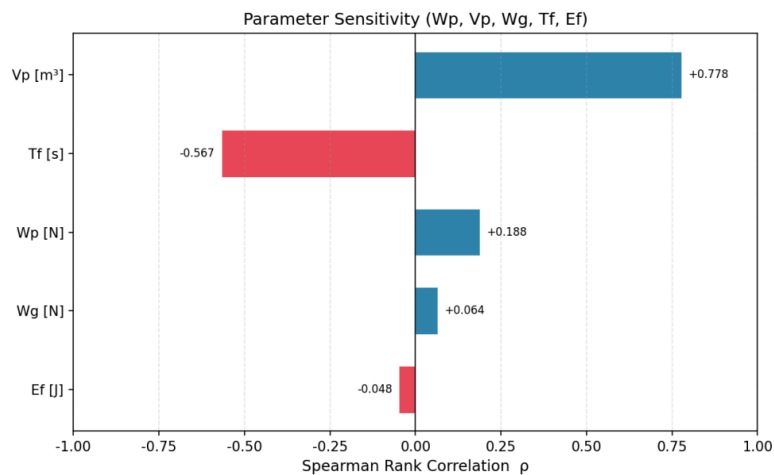
Manufacturing focuses heavily on minimizing overall system weight while maintaining system reliability. The bay doors are laser-cut from balsa wood and wrapped in monokote. The bay doors are sized with margins relative to the volume payload to ensure sufficient clearance for deployment. Laser-cut servo mounts slide directly into the front and rear firewalls, ensuring a rigid connection. A CAD design of the release mechanism can be seen in Appendix 8.2.3.

8. References and Appendices

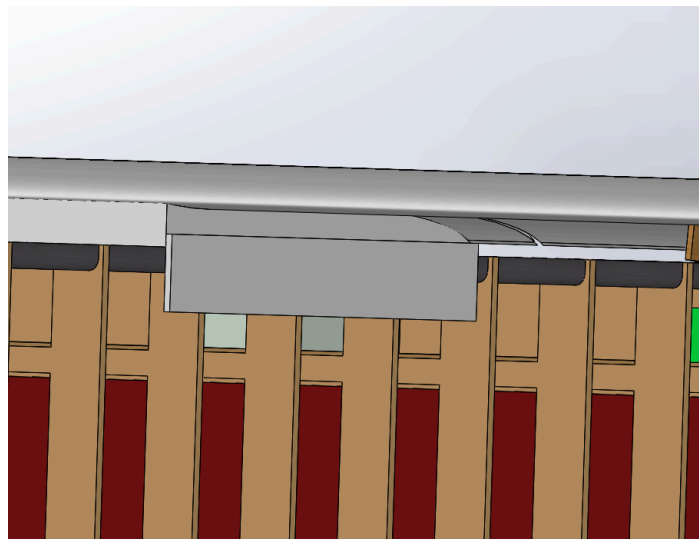
8.1 References

1. John T. Hwang, *MAE 155B Lecture: 01-Introduction*, 2026
2. Raymer, D. P., *Aircraft Design: A Conceptual Approach*, 6th ed., American Institute of Aeronautics and Astronautics, Reston, VA, 2018.
3. APC Propellers, *Performance Data*, APC Propellers, <https://www.apcprop.com/technical-information/performance-data/>, accessed April 21, 2026.

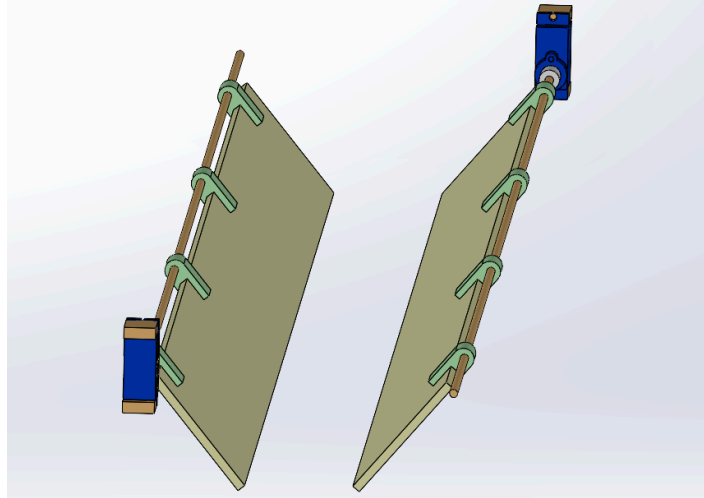
8.2 Appendices



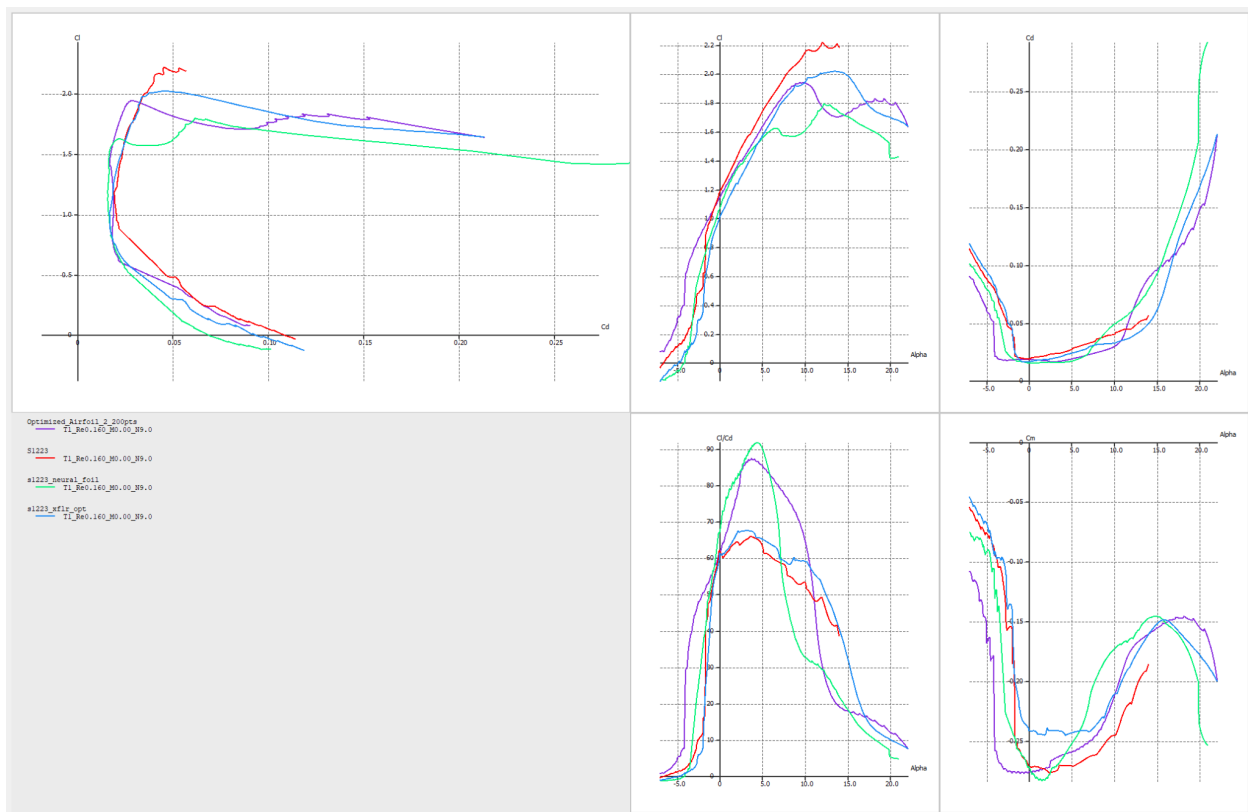
Appendix 8.2.1: Major influences of scoring parameters on profitability resulting from the Monte Carlo analysis in section 2.



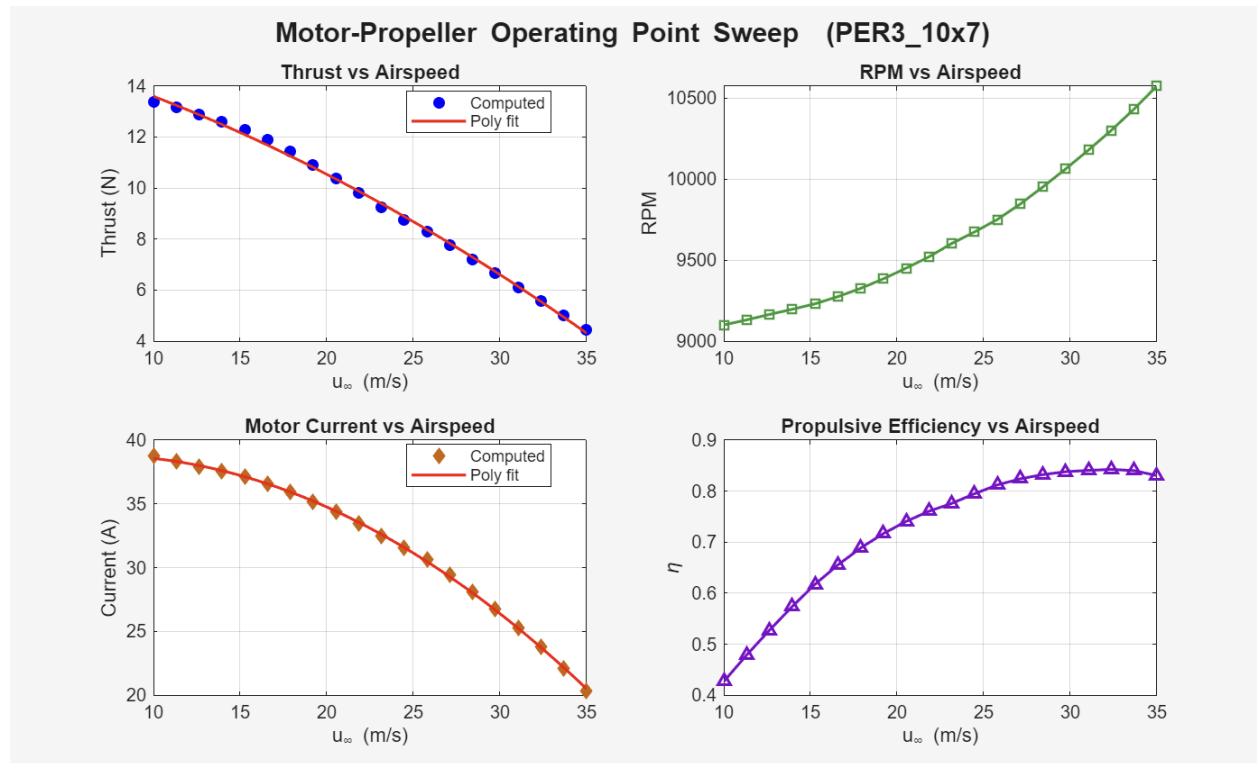
Appendix 8.2.2: Computer-Aided-Design of wing saddle mounted on the fuselage



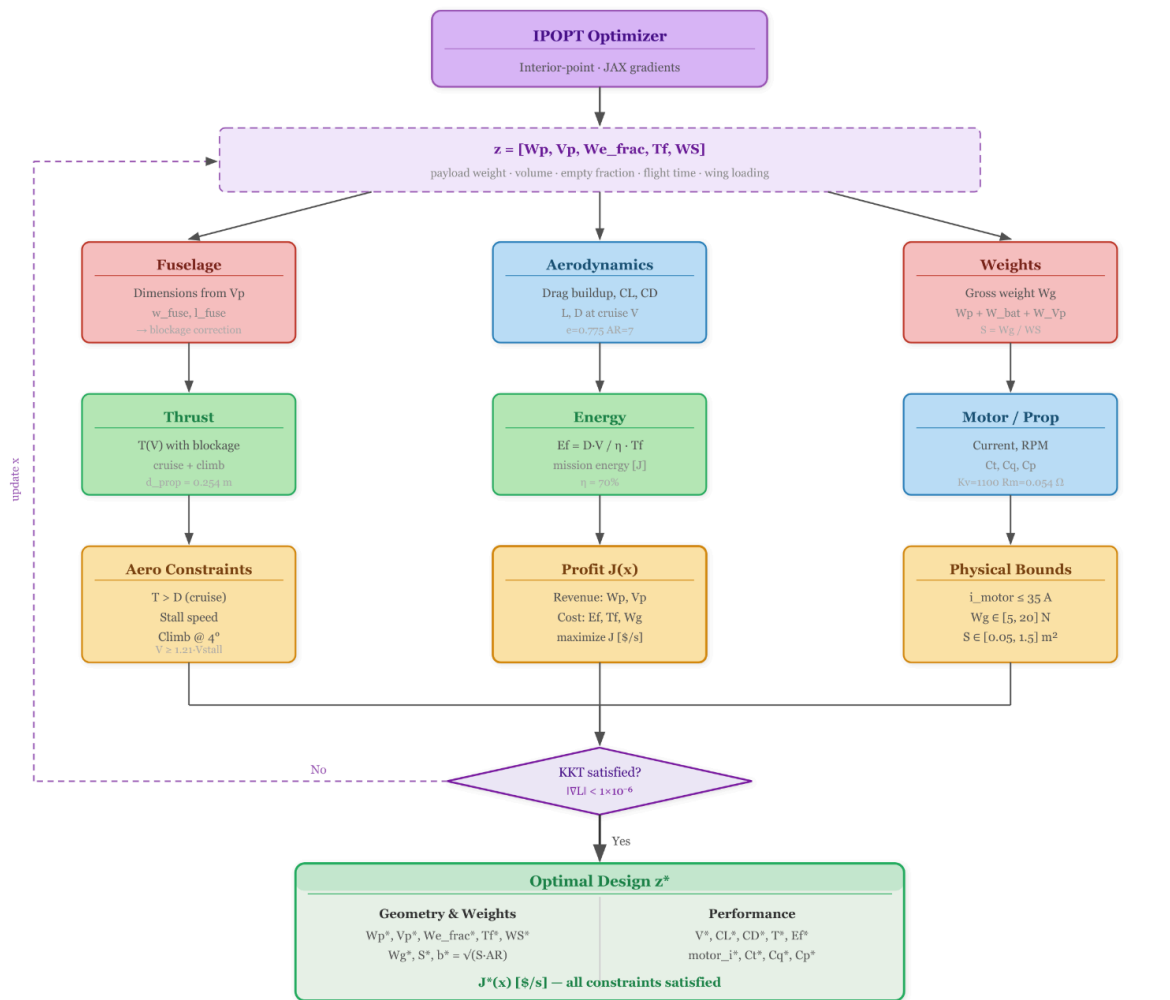
Appendix 8.2.3: Computer-Aided-Design of the Twin Bay Door Mechanism (Open)



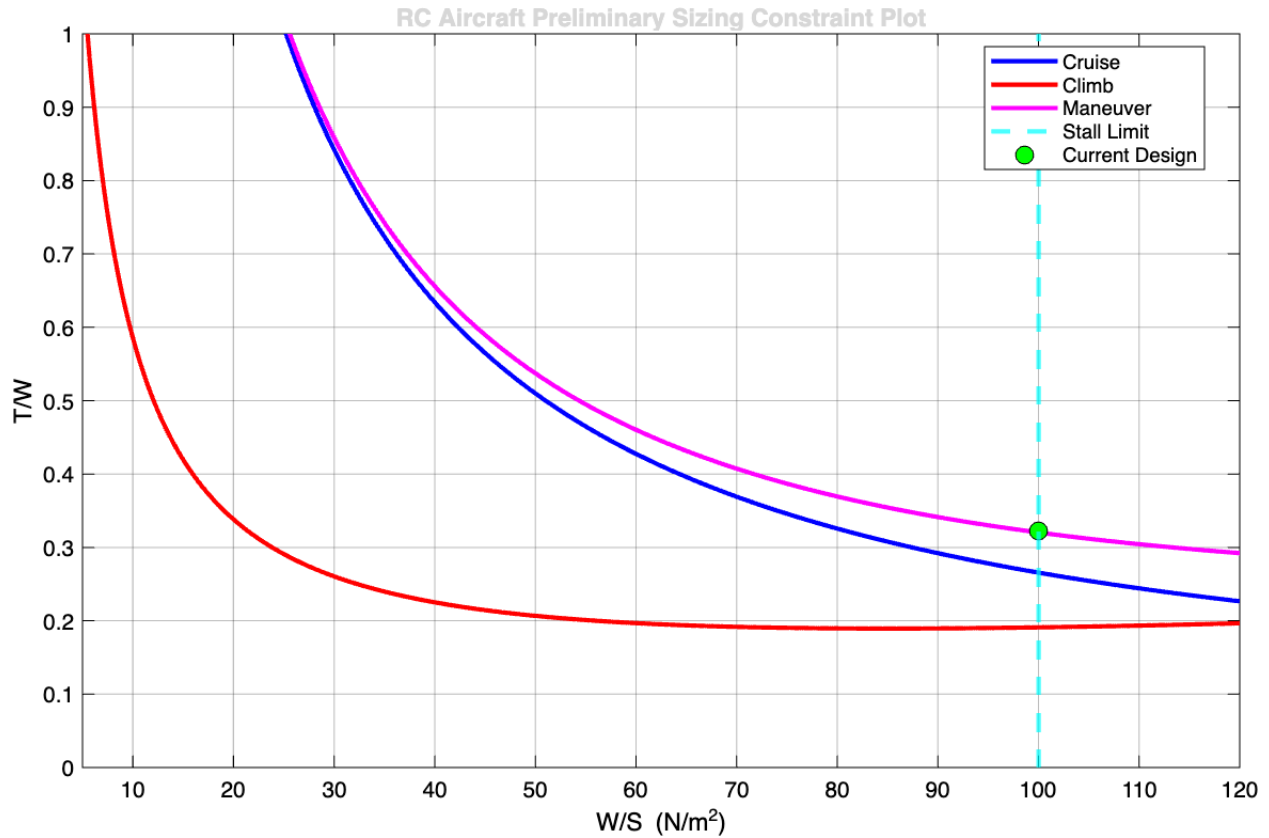
Appendix 8.2.4: XFLR5 2D Aerodynamic analysis of candidate airfoils for a Reynold's number of 184,000. The standard S1223 airfoil in red, the XFLR5-optimized airfoil in blue, the first Neural Foil optimized airfoil (for small range of angles of attack) in green, and the selected S1223 Neural Foil optimized airfoil in purple. From left to right: C_l vs C_d graph, C_l vs α graph, C_d vs α graph, C_l/C_d vs α graph, C_m vs α graph.



Appendix 8.2.5: Various Propulsion Characteristics Plotted Against Cruise Speed



Appendix 8.2.6: Overview of gradient-based MDO workflow implemented in section 2.

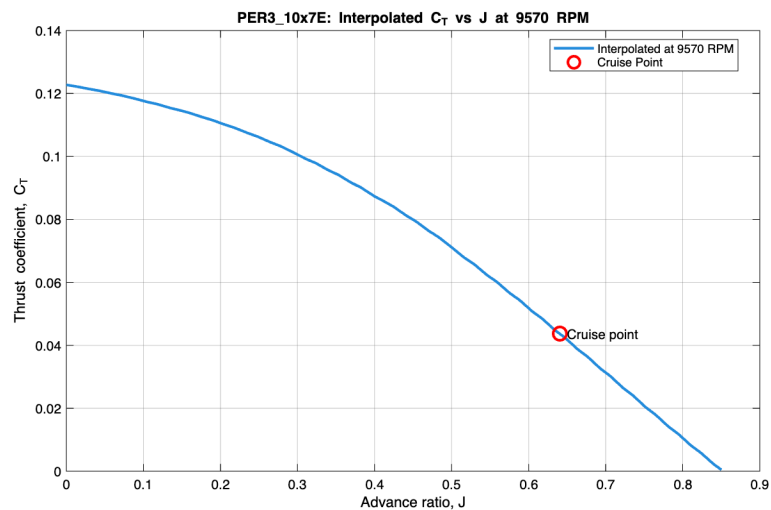


Appendix 8.2.7: Sizing carpet plot T/W vs W/S

Appendix 8.2.8: Table of main propulsion parameters for the 10x7E propeller

Propeller	RPM [rev / min]	Thrust Coefficient	Torque Coefficient	Efficiency
APC 10x7E	9570	0.082	0.01	0.68

Appendix 8.2.9: Graph of Thrust Coefficient vs. Advance Ratio for 10x7E prop with the operating point highlighted.



8.3: Components, Materials, and Dimensions

8.3.1: XPS Foam Wing: 1184 x 168 x 20 mm

8.3.2: Flat Carbon Fiber Spar: 1 x 6 x 1000 mm

8.3.3 SMRAZA SG90 9g Micro Servo: 22.2 x 11.8 x 31 mm

8.3.4 Plywood Servo Mount: 22.2 x 11.8 x 31 mm

8.3.5 Foam Vertical Stabilizer: 77 x 8.5 x 117mm

8.3.6 Foam Horizontal Stabilizer: 309 x 77 x 10 mm

8.3.7 3D-Printed PLA Nose Cone: base: 157 x 163 mm, tip diameter: 35 mm

8.3.8 Wooden rib cutouts: 163 x 157 x 3 mm

8.3.9 Carbon Fiber Spar: 10mm outer diameter, 2.3mm thick, 1000mm long

8.3.10 Wooden Spar: 1000 x 10 x 10 mm

8.3.11 Firewall: 163 x 3 x 157 mm

8.3.12 Payload Bay Door: 52 x 3 x 710 mm

8.3.13 Wooden payload rod: 3mm diameter x 760 mm

8.3.14 Payload mechanism hinge: 8 x 15 x 19 mm